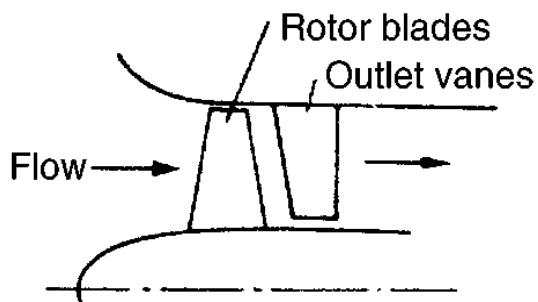
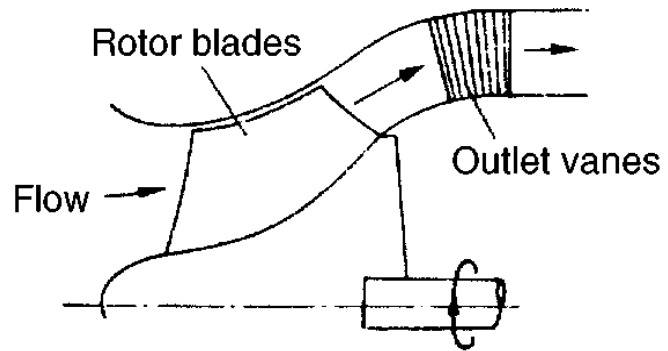


Definition of a turbo machine

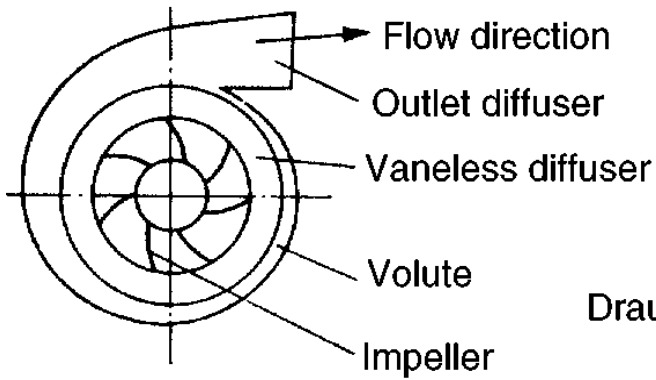
- ❖ We classify as turbo machines all those devices in which energy is transferred either to, or from, a continuously flowing fluid by the *dynamic action* of one or more moving blade rows.
- ❖ The word *turbo* or *turbinis* is of Latin origin and implies that which spins or whirls around. Essentially, a rotating blade row, a *rotor* or an *impeller* changes the stagnation enthalpy of the fluid moving through it by either doing positive or negative work, depending upon the effect required of the machine.
- ❖ These enthalpy changes are intimately linked with the pressure changes occurring simultaneously in the fluid.
- ❖ The definition of a turbo machine as stated above, is rather too general for the purposes of this book as it embraces *open* turbo machines such as propellers, wind turbines and enshrouded fans, all of which influence the state of a not readily quantifiable flow of a fluid.
- ❖ The subject *fluid mechanics, thermodynamics of turbo machinery*, therefore, is limited to machines enclosed by a closely fitting casing or shroud through which a readily measurable quantity of fluid passes in unit time.
- ❖ The subject of open turbo machines is covered by the classic text of Glauert (1959) or by Duncan *et al.* (1970), the elementary treatment of propellers by general fluid mechanics textbooks such as Streeter and Wylie (1979) or Massey (1979), and the important, still developing subject of wind turbines, by Freris (1990).
- ❖ Two main categories of turbo machine are identified: firstly, those which *absorb* power to increase the fluid pressure or head (ducted fans, compressors and pumps); secondly, those that *produce* power by expanding fluid to a lower pressure or head (hydraulic, steam and gas turbines).
- ❖ Figure shows, in a simple diagrammatic form, a selection of the many different varieties of turbo machine encountered in practice. The reason that so many different types of either pump (compressor) or turbine are in use is because of the almost infinite range of service requirements.
- ❖ Generally speaking, for a given set of operating requirements there is one type of pump or turbine best suited to provide optimum conditions of operation. This point is discussed more fully in the section of this chapter concerned with specific speed.
- ❖ Turbo machines are further categorised according to the nature of the flow path through the passages of the rotor. When the path of the *through-flow* is wholly or mainly parallel to the axis of rotation, the device is termed an *axial flow turbo machine*
- ❖ Figure (a) and (e) when the path of the *through-flow* is wholly or mainly in a plane perpendicular to the rotation axis, the device is termed a *radial flow turbo machine* (e.g. Figure (c)). *Mixed flow turbo machines* are widely used.
- ❖ The term *mixed flow* in this context refers to the direction of the through-flow at rotor outlet when both radial and axial velocity components are present in significant amounts. Figure (b) shows a mixed flow pump and Figure (d) a mixed flow hydraulic turbine.
- ❖ One further category should be mentioned. All turbo machines can be classified as either *impulse* or *reaction* machines according to whether pressure changes are absent or present respectively in the flow through the rotor.
- ❖ In an impulse machine all the pressure change takes place in one or more nozzles, the fluid being directed onto the rotor. The Pelton wheel, Figure (f), is an example of an impulse turbine.
- ❖ The main purpose of this book is to examine, through the laws of fluid mechanics and thermodynamics, the means by which the energy transfer is achieved in the chief types of turbo machine, together with the differing behaviour of individual types in operation.
- ❖ Methods of analysing the flow processes differ depending upon the geometrical configuration of the machine, on whether the fluid can be regarded as incompressible or not and whether the machine absorbs or produces work.
- ❖ As far as possible, a unified treatment is adopted so that machines having similar configurations and function are considered together.



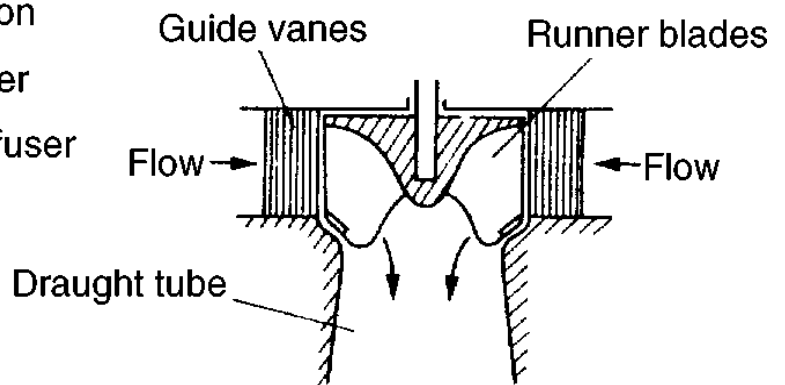
(a) Single stage axial flow compressor or pump



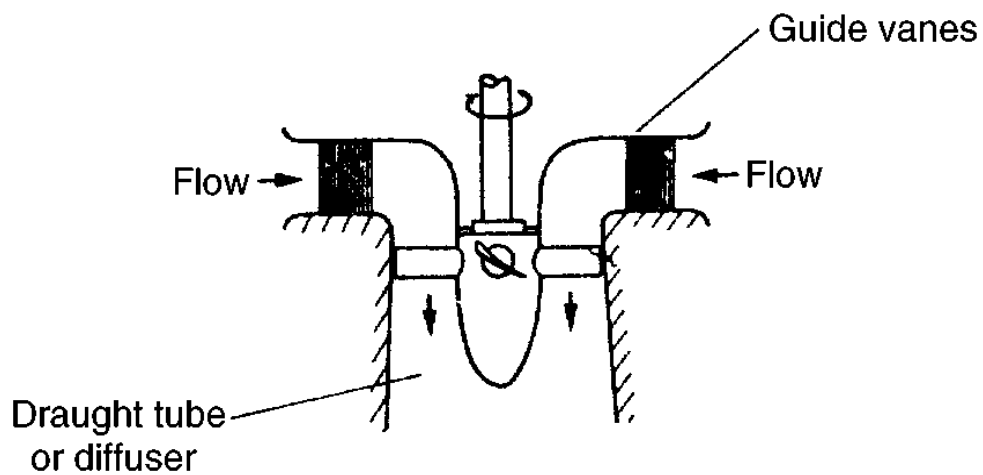
(b) Mixed flow pump



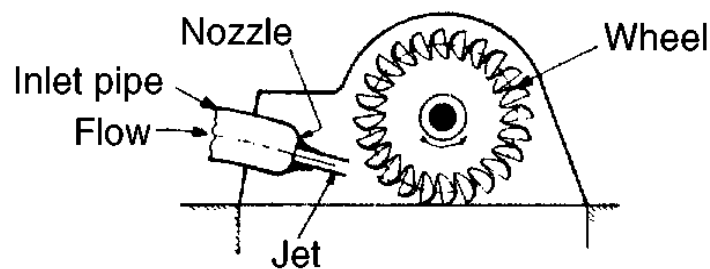
(c) Centrifugal compressor or pump



(d) Francis turbine (mixed flow type)



(e) Kaplan turbine



(f) Pelton wheel

IMPACTS OF WATER JETS

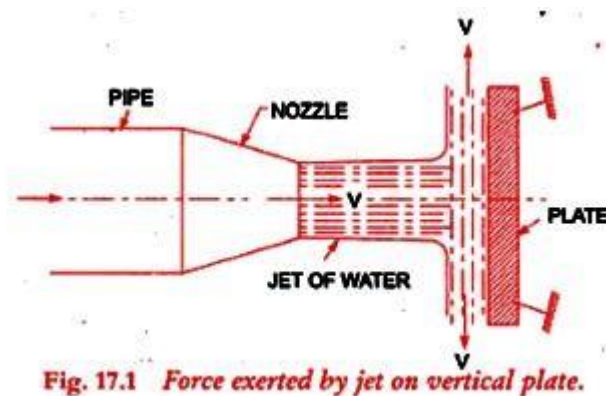
INTRODUCTION

The liquid comes out in the form of a jet from the outlet of the nozzle, which is fitted to a pipe through which the liquid is flowing under pressure. This force is obtained from Newton's second law of motion or from impulse momentum equation. Thus impact of jet means force exerted by the jet on a plate which may be stationary or moving.

1. Force exerted by the jet on a stationary plate when
 - a) Plate is vertical to the jet,
 - b) Plate is inclined to the jet,
 - c) Plate is curved.
2. Force exerted by the jet on a moving plate, when
 - a) Plate is vertical to the jet,
 - b) Plate is inclined to the jet,
 - c) Plate is curved
3. Force exerted by the jet on Hinged plate

FORCE IS EXERTED BY A JET ON A STATIONARY VERTICAL PLATE

Consider a jet of water coming out from the nozzle, strikes a flat vertical plate as shown in figure.



Let V = velocity of the jet, d = diameter of the jet, a = area of cross section of the jet = –

Initial velocity in the direction of X axis = V

Final velocity in the direction of X axis = 0

Initial velocity in the Direction of Y axis = 0

Final velocity in the Direction of Y axis = 0

F = Rate of change of momentum in the direction of force

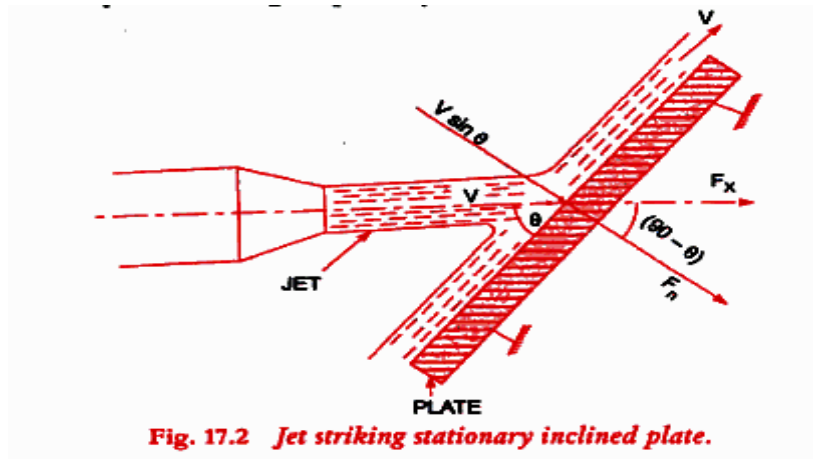
=

F_x =

$F_y = 0$

FORCE IS EXERTED BY A JET ON A STATIONARY INCLINED FLAT PLATE

Consider a jet of water coming out from the nozzle, strikes an inclined flat plate as shown in figure.



V = velocity of the jet in the direction of x , a = area of cross section of the jet

Initial

velocity in the direction of X axis = V Final velocity in

the direction of X axis = 0

Initial velocity in the Direction of Y axis = 0

Final velocity in the Direction of Y axis = 0

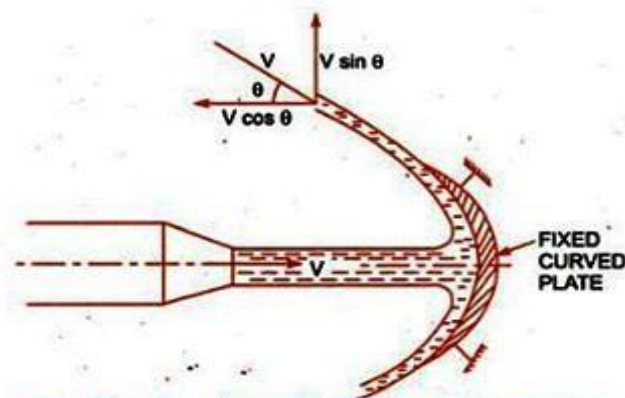
Mass of water per second =

F_x =

F_y =

FORCE IS EXERTED BY A JET ON A STATIONARY CURVED PLATE

(A) At the Centre



Initial velocity in the direction of X axis = V

Final velocity in the direction of X axis = $-V$

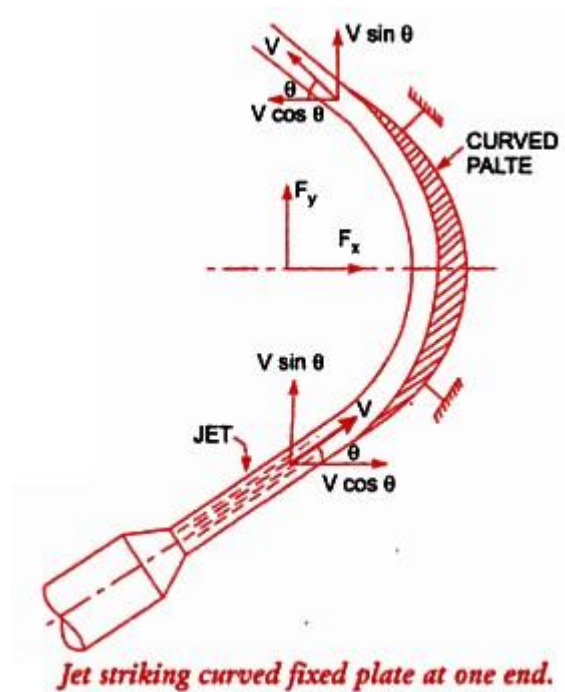
Initial velocity in the Direction of Y axis = 0

Final velocity in the Direction of Y axis = V

F_x =

F_y =

(B) At one end tangential when the plate is symmetrical



Initial velocity in the direction of X axis = V

Final velocity in the direction of X axis = $-V$

Initial velocity in the Direction of Y axis = V

Final velocity in the Direction of Y axis = V

$$F_x =$$

$$F_y = 0$$

(c) At one end tangentially when the plate is unsymmetrical

Initial velocity in the direction of X axis = V

Final velocity in the direction of X axis = $-V$

Initial velocity in the Direction of Y axis = V

Final velocity in the Direction of Y axis = V

$$F_x =$$

$$F_y =$$

FORCE EXERTED BY A JET ON A HINGED PLATE

X = distance of the centre of jet from hinge O,

W = weight of plate acting at centre of gravity of the plate.

1. Force due to jet of water, normal to the plate,

$$F_n =$$

2. Weight of the plate, W

$$W = \text{---}$$

FORCE ON A FLATE VERTICAL PLATE MOVING IN THE DIRECTION OF THE JET

V = velocity of jet (absolute)

a = Area of cross section of the jet,

u = Velocity of flat plate.

Mass of water per second =

Initial velocity in the direction of X axis = $[V - u]$

Final velocity in the direction of X axis = 0

Initial velocity in the Direction of Y axis = 0

Final velocity in the Direction of Y axis = 0

$$F_x =$$

$$F_y = 0$$

WORK DONE PER SECOND BY THE JET ON THE PLATE

FORCE ON A INCLINED PLATE MOVING IN THE DIRECTION OF THE JET

Initial velocity in the direction of X axis = $[V - u]$

Final velocity in the direction of X axis = 0

Initial velocity in the Direction of Y axis = $[V - u]$

Final velocity in the Direction of Y axis = 0

$$F_x =$$

$$F_y =$$

WORK DONE PER SECOND BY THE JET ON THE PLATE

FORCE ON A CURVED PLATE MOVING IN THE DIRECTION OF THE JET

Initial velocity in the direction of X axis = $[V - u]$

Final velocity in the direction of X axis = $- [V - u]$

Initial velocity in the Direction of Y axis = 0

Final velocity in the Direction of Y axis = $[V - u]$

$$F_x =$$

$$F_y =$$

WORK DONE PER SECOND BY THE JET ON THE PLATE

FROCE EXERTED ON A JET OF WATER ON AN UNSYMMETRICAL MOVING CURVED PLATE WHEN JET STRIKES TANGENTIALLY AT ONE OF THE TIPS

Let

V_1 = Velocity of jet at inlet

U_1 = velocity of plate at inlet

V_{r1} = Relative velocity of jet and plate at inlet

V_{w1} = velocity of whirl at inlet

V_{f1} = velocity of flow at inlet

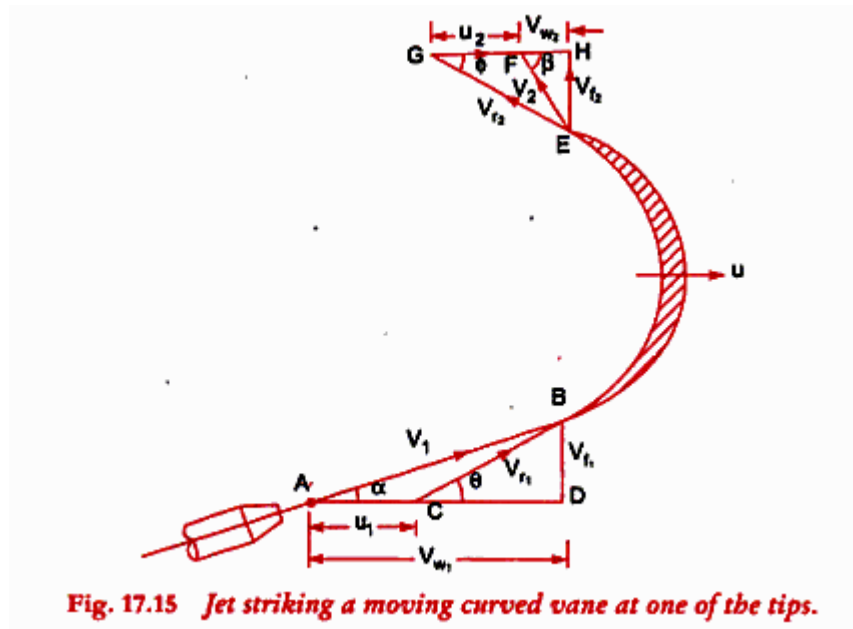
V_2 = Velocity of jet at outlet

U_2 = velocity of plate at outlet

V_{r2} = Relative velocity of jet and plate at outlet

V_{w2} = velocity of whirl at outlet

V_{f2} = velocity of flow at outlet



Force exerted by the jet in the direction of motion

$$F_x = aV_{r1} [V_{w1} \pm V_{w2}]$$

When is acute angle

$$F_x = aV_{r1} [V_{w1} + V_{w2}]$$

When is obtuse angle

$$F_x = aV_{r1} [V_{w1} - V_{w2}]$$

When is right angle

$$F_x = a V_{r1} [V_{w1}]$$

WORK DONE PER SECOND BY THE JET ON THE VANE

$$= a V_{r1} [V_{w1} \pm V_{w2}] \times u$$

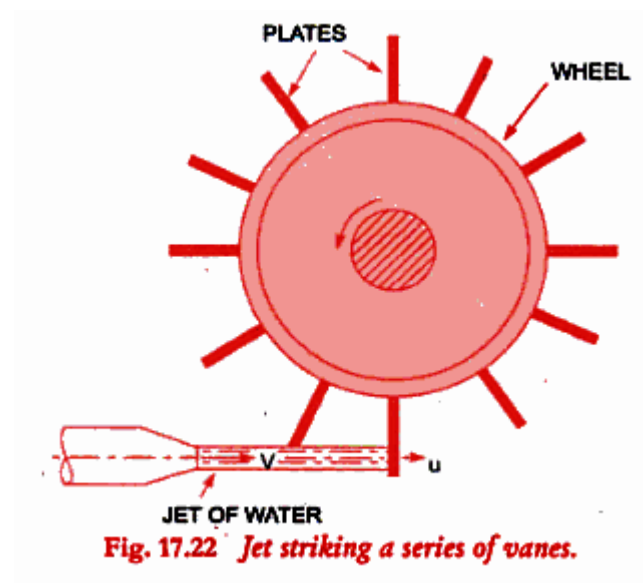
SECOND PER WORK DONE PER UNIT WEIGHT OF FLUID STRIKING PER SECOND

$$= - [V_{w1} \pm V_{w2}] \times u \quad \text{Nm/N}$$

SECOND PER WORK DONE PER UNIT MASS OF FLUID STRIKING PER SECOND

$$= [V_{w1} \pm V_{w2}] \times u \quad \text{Nm/N}$$

FORCE EXERTED BY A JET OF WATER ON A SERIES OF VANES



V = velocity of jet (absolute)

a = Area of cross section of the jet,

u = Velocity of vane.

Mass of water per second =

Initial velocity in the direction of X axis = $[V - u]$

Final velocity in the direction of X axis = 0

Initial velocity in the Direction of Y axis = 0

Final velocity in the Direction of Y axis = 0

$$F_x =$$

$$F_y = 0$$

WORK DONE PER SECOND BY THE JET ON THE PLATE

Kinetic energy of jet per second

– –

– V^3

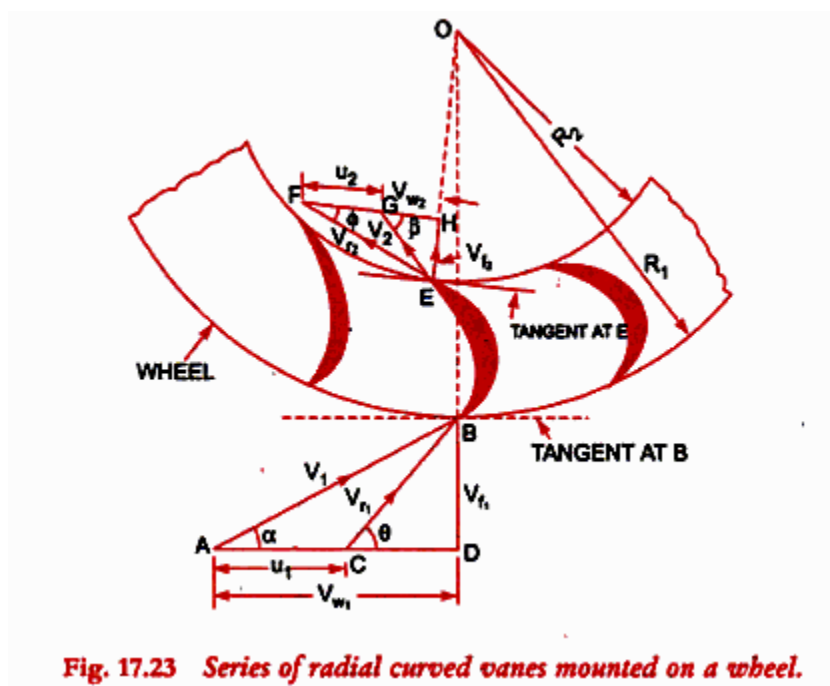
Efficiency = _____

Condition for maximum efficiency,

–

Maximum Efficiency = 0.5 = 50 %

FORCE EXERTED BY A JET OF WATER ON A SERIES OF RADIAL CURVED VANES



R_1 = Radius of Wheel at inlet of the vane,

R_2 = Radius of Wheel at outlet of the vane,

The mass of water striking per second for a series of vanes = aV_1

Momentum of water striking the Vanes in tangential direction per second = $aV_1 \times V_{w1}$

$$V_{w1} = V_1$$

Similarly Momentum of water at outlet per second = $-aV_1 \times V_{w2}$

$$V_{w2} = V_2$$

Angular Momentum at Inlet = Momentum at Inlet x Radius at Inlet

$$= aV_1 \times V_{w1} \times R_1$$

Angular Momentum at outlet = Momentum at outlet x Radius at outlet

$$= -aV_1 \times V_{w2} \times R_2$$

Torque exerted by the water on the wheel,

T = rate of change of angular Momentum

= [Initial angular momentum per second – final angular momentum per second]

$$= aV_1 \times V_{w1} \times R_1 - (-aV_1 \times V_{w2} \times R_2)$$

$$= aV_1 [V_{w1} \times R_1 + V_{w2} R_2]$$

Work done per second on the wheel = Torque x Angular Velocity = $T \times$

$$= aV_1 [V_{w1} \times R_1 + V_{w2} R_2] \times$$

$$= aV_1 [V_{w1} \times U_1 + V_{w2} U_2] \quad (,)$$

If the angle is an obtuse angle in the figure then the work done per second will be given as

$$= aV_1 [V_{w1} \times U_1 - V_{w2} U_2]$$

The general expression for the work done per second on the wheel

$$= aV_1 [V_{w1} \times U_1 \pm V_{w2} U_2]$$

If the discharge is radial at outlet, then $= 90^\circ$ and work done becomes

$$= aV_1 [V_{w1} \times U_1]$$

EFFICIENCY OF THE RADIAL CURVED VANES

$$\text{Efficiency} = \frac{\text{Work done on the wheel}}{\text{Work done by the water}}$$

$$= \frac{aV_1 [V_{w1} \times U_1]}{aV_1 [V_{w1} \times V_1]}$$

1. A jet of water having a velocity of 20 m/s strikes a curved vane, which is moving with a velocity of 10 m/s. The jet makes an angle of 20° with the direction of motion of vane at inlet and leaves at **an angle of 130° to the direction of motion of vane an outlet**. Calculate :

- (i) Vane angles, so that the water enters and leaves the vane without shock.
(ii) Work done per second per unit weight of water striking the vane per second

Given:

$$V_1 = 20 \text{ m/s}$$

$$u_1 = 10 \text{ m/s}$$

$$= 20^\circ$$

$$= 180^\circ - 130^\circ = 50^\circ$$

$$u_1 = u_2$$

$$V_{r1} = V_{r2}$$

$$V_{f1} = 6.84 \text{ m/s}$$

$$= \text{---}$$

$$V_{w1} = 18.794 \text{ m/s}$$

$$= \text{---}$$

$$= 37^\circ 52.5'$$

$$= \text{---}$$

To Find:

$$=?$$

$$=?$$

$$\text{W.D per unit weight} = ?$$

$$V_{r1} = V_{r2} = 11.14 \text{ m/s}$$

b. From outlet Velocity triangle

Applying sine rule

$$\text{---} = \text{---}$$

$$y = 180 - = 130^\circ$$

$$x = 43^\circ 26'$$

$$x + y + = 180^\circ$$

$$= 6^\circ 33'$$

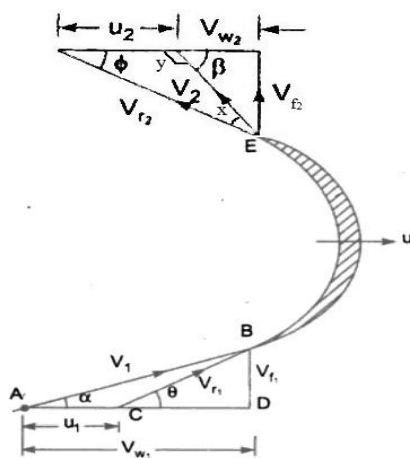
c. Work done per unit weight:

$$= \text{---}$$

$$V_{w2} = 1.067 \text{ m/s}$$

$$\text{W.D} = \text{---}$$

$$\text{W.D} = 20.24 \text{ Nm/N}$$



a. From inlet velocity triangle

$$= \text{---}$$

2. A jet of water having a velocity of 40 m/s strikes a curved vane, which is moving with a velocity of 20 m/s. The jet makes an angle of 30° with the direction of motion of vane at inlet and leaves at **an angle of 90° to the direction of motion of vane outlet**. Draw the velocity triangles at inlet and outlet determine the vane angles at inlet and outlet so that the water enters and leaves the vane without shock.

Given:

$$V_1 = 40 \text{ m/s}$$

$$u_1 = 20 \text{ m/s}$$

$$= 30^\circ$$

$$= 180^\circ - 90^\circ = 90^\circ$$

$$u_1 = u_2$$

$$V_{r1} = V_{r2}$$

d. From inlet velocity triangle

$$= \text{---}$$

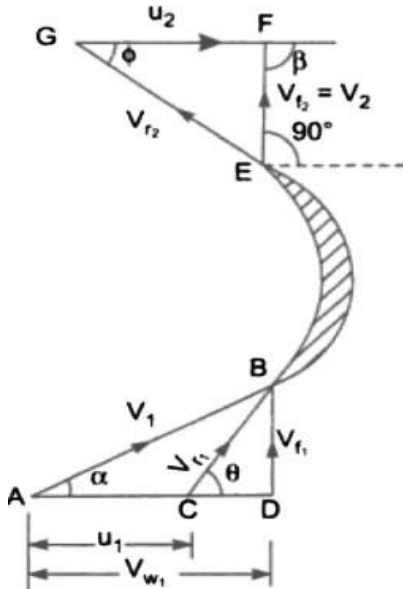
$$=?$$

$$V_{f1} = 20 \text{ m/s}$$

$$=?$$

$$= \text{---}$$

Solution:



$$V_{w1} = 34.64 \text{ m/s}$$

$$= \text{---}$$

$$= 53^\circ 47.4'$$

$$= \text{---}$$

$$V_{r1} = V_{r2} = 24.78 \text{ m/s}$$

e. From outlet Velocity triangle

$$= \text{---}$$

$$= 36^\circ 10'$$

3. A jet of water of diameter 50 mm, having a velocity of 20 m/s strikes a curved vane which is moving with a velocity of 10 m/s in **the direction of the jet**. The jet leaves the vane of an **angle of 60° to the direction of motion of vane at outlet**. Determine:

- The force exerted by the jet on the vane in the direction of motion
- Work done per second of the jet

Given:

$$d = 0.05 \text{ m}$$

$$V_1 = 20 \text{ m/s}$$

$$u_1 = 10 \text{ m/s}$$

$$= 0^\circ$$

$$= 180^\circ - 60^\circ = 120^\circ$$

$$u_1 = u_2$$

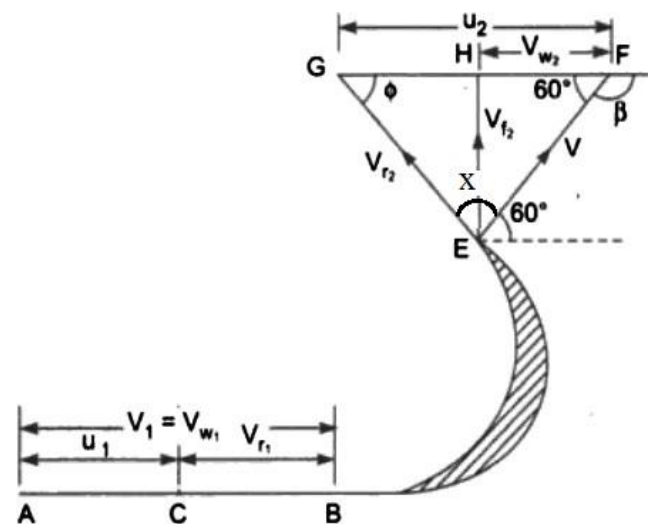
$$V_{r1} = V_{r2}$$

To Find:

Force exerted by the jet = ?

W.D per second = ?

Solution:



From inlet velocity triangle

$$V_{r1} = V_1 - u_1 = 10 \text{ m/s}$$

$$V_{w1} = V_1 = 20 \text{ m/s}$$

$$V_{r2} = V_{r1} = 10 \text{ m/s}$$

From outlet velocity triangle

$$\text{---} = \text{---}$$

$$V_{w2} = 5 \text{ m/s}$$

$$x = 60^\circ$$

$$x + 60^\circ = 180^\circ$$

$$= 60^\circ$$

$$= \text{---}$$

a. Force exerted by the jet:

$$F_x = a V_{r1} \text{ ---}$$

$$F_x = 294.45 \text{ N}$$

b. Work done per second:

$$W.D = a V_{r1} \text{ ---} \times u$$

$$W.D = 2944.5 \text{ Nm/s}$$

4. A jet of water having a velocity of 15 m/s strikes a curved vane which is moving with a velocity of 5 m/s. The vane symmetrical and is so shaped that **the jet is deflected through 120°**. Find the angle of the jet at inlet of the vane so that there is no shock. What is the absolute velocity of the jet at outlet in magnitude and direction and work done per unit weight of water? Assume the vane to be smooth.

Given:

$$V_1 = 20 \text{ m/s}$$

$$u_1 = 10 \text{ m/s}$$

$$= 180^\circ - 120^\circ = 60^\circ$$

$$= 30^\circ$$

$$= 30^\circ$$

$$u_1 = u_2, V_{r1} = V_{r2}$$

From inlet velocity triangle

Applying sine rule

$$\text{---} = \text{---}$$

$$p = 180^\circ - 30^\circ = 150^\circ$$

$$q = 9.596^\circ$$

$$p + q = 180^\circ$$

$$= 20.24^\circ$$

Applying sine rule

$$\text{---} = \text{---}$$

$$V_{r1} = V_{r2} = 10.46 \text{ m/s}$$

From outlet velocity triangle

$$\text{---} = V_{w2} = 4.06 \text{ m/s}$$

$$= V_{f2} = 5.23 \text{ m/s}$$

$$V_2 = (V_{f2}^2 + V_{w2}^2)^{1/2} = 6.62 \text{ m/s}$$

$$= \text{---}$$

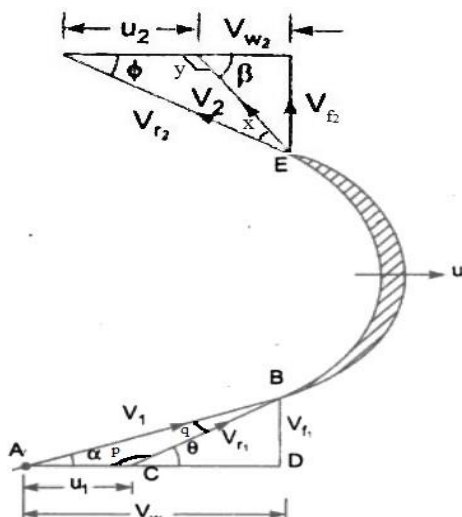
$$= 52^\circ 10'$$

$$* = 180^\circ - 52^\circ 10' = 127^\circ 50'$$

Work done per unit weight:

$$W.D = \text{---}$$

$$W.D = 9.225 \text{ Nm/N}$$



5. A jet of water moving at 12 m/s impinges on vane shaped to **deflect the jet through 120°** when stationary. If the vane is moving at 5 m/s, find the angle of the jet so that there is no shock at inlet. What is the absolute velocity of the jet at exit in magnitude and direction and the work done per second per unit weight of water striking per second? Assume that the vane is smooth.

Given:

$$V_1 = 12 \text{ m/s}$$

$$u_1 = 5 \text{ m/s}$$

$$= 180^\circ - 120^\circ = 60^\circ$$

$$= 30^\circ$$

$$= 30^\circ$$

$$u_1 = u_2, V_{r1} = V_{r2}$$

To Find:

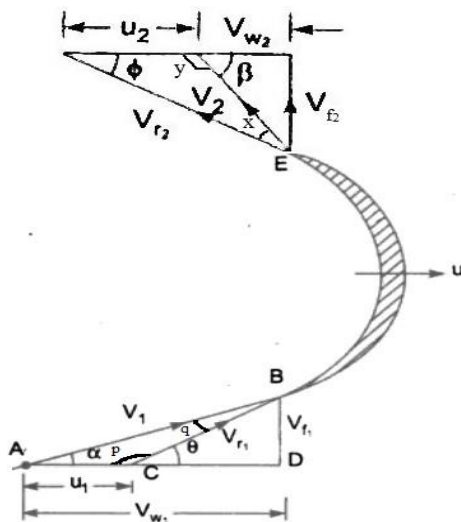
$$=?$$

$$V_2 = ?$$

$$* = ?$$

$$\text{W.D per unit weight} = ?$$

Solution:



From inlet velocity triangle

Applying sine rule

$$\frac{\sin p}{V_{r1}} = \frac{\sin q}{u_1}$$

$$p = 180^\circ - 30^\circ = 150^\circ$$

$$q = 12.02^\circ$$

$$p + q + = 180^\circ$$

$$= 17.59'$$

Applying sine rule

$$\frac{\sin p}{V_{r1}} = \frac{\sin q}{u_1}$$

$$V_{r1} = V_{r2} = 7.41 \text{ m/s}$$

From outlet velocity triangle

$$\frac{\sin p}{V_{r2}} = \frac{\sin q}{u_2} \Rightarrow V_{r2} = 1.417 \text{ m/s}$$

$$V_2 = \sqrt{V_{r2}^2 + V_{w2}^2} = 3.705 \text{ m/s}$$

$$V_2 = \sqrt{V_{f2}^2 + V_{w2}^2} = 3.96 \text{ m/s}$$

$$= 69.42'$$

$$* = 180 - = 110.55.8'$$

Work done per unit weight:

$$\text{W.D} = \frac{V_{w1} - V_{w2}}{g}$$

$$\text{W.D} = 6.537 \text{ Nm/N}$$

6. A jet of water having a velocity of 15 m/s, strikes a curved vane which is moving with a velocity of 5 m/s **in the same direction as that of jet at inlet**. The vane is so shaped that **the jet is deflected through 135°**. The diameter of jet is 100 mm. Assuming the vane to be smooth, find: (i) Force exerted by the jet on the vane in the direction of motion, (ii) Power exerted on the vane, (iii) Efficiency of the vane.

Given:

$$V_1 = 15 \text{ m/s}$$

$$u_1 = 5 \text{ m/s}$$

$$= 180^\circ - 135^\circ = 45^\circ$$

$$= 45^\circ$$

$$u_1 = u_2, V_{r1} = V_{r2}$$

To Find:

Force exerted by the jet =?

Power exerted by the vane =?

Efficiency of the vane =?

Solution:

From inlet velocity triangle

$$V_{r1} = V_1 - u_1 = 10 \text{ m/s}$$

$$V_{w1} = V_1 = 15 \text{ m/s}$$

$$V_{r2} = V_{r1} = 10 \text{ m/s}$$

From outlet velocity triangle

$$= \text{---}$$

$$V_{w2} = 2.07 \text{ m/s}$$

a. Force exerted by the jet

$$F_X = \text{---} \times a V_{r1}$$

$$F_X = 1340.6 \text{ N}$$

b. Power of the vane

$$P = \text{---} \times 6.703 \text{ kW}$$

c. Efficiency:

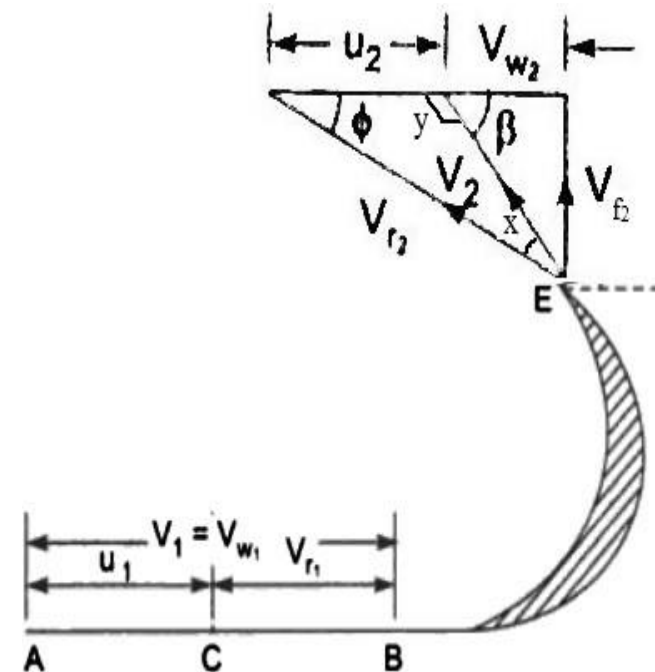
$$\text{Efficiency} = \text{---}$$

$$W.D = \text{---} \times u = 6703 \text{ Nm/s}$$

$$K.E = \text{---} \times V_1^3 \text{---}$$

$$K.E = 13253.6 \text{ Nm/s}$$

$$\text{Efficiency} = 50.5 \%$$



7. A jet of water having a velocity of 35 m/s impinges on a series of vanes moving with a velocity of 20 m/s. The jet makes an **angle of 30° to the direction of motion of vanes when entering and leaves at an angle of 120°**. Draw the velocity triangles and find: (a) the angles of vane tips so that water enters and leaves without shock, (b) the work done per unit weight of water entering the vanes (c) the efficiency

Given:

$$V_1 = 35 \text{ m/s}$$

$$u_1 = 20 \text{ m/s}$$

$$= 30^\circ$$

$$= 180^\circ - 120^\circ = 60^\circ$$

$$u_1 = u_2$$

$$V_{r1} = V_{r2}$$

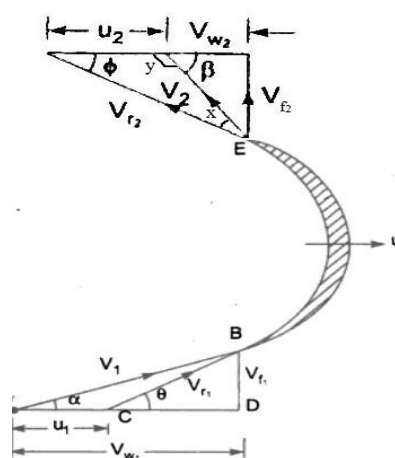
To Find:

$$=?$$

$$=?$$

W.D per unit weight =?

Efficiency of the vane =?

Solution:

From inlet velocity triangle

$$= \text{---}$$

$$V_{f1} = 17.50 \text{ m/s}$$

$$= \text{---}$$

$$V_{w1} = 30.31 \text{ m/s}$$

$$= \text{---}$$

$$= 60^\circ$$

$$= \text{---}$$

$$V_{r1} = V_{r2} = 20.25 \text{ m/s}$$

From outlet Velocity triangle

Applying sine rule

$$\text{---} = \text{---}$$

$$y = 180 - = 120^\circ$$

$$x = 58.75^\circ$$

$$x + y + = 180^\circ$$

$$= 1.25^\circ$$

Work done per unit weight:

$$= \text{---}$$

$$V_{w2} = 0.24 \text{ m/s}$$

$$W.D = \text{---}$$

$$W.D = 62.28 \text{ Nm/N}$$

Efficiency:

$$\text{Efficiency} = \text{---}$$

$$K.E = \text{---} \text{ (per unit weight } m = 1/g)$$

$$K.E = 62.43$$

$$\text{Efficiency} = 99.74 \%$$

8. A jet of water having a velocity of 30 m/s strikes a series of radial curved vanes mounted on a wheel which is rotating at 200 r.p.m. The jet makes an angle of 20° with tangent to the wheel at inlet and leaves the wheel with a velocity of 5 m/s at an angle of 130° to the tangent to the wheel at outlet. **Water is flowing outward in radial direction.** The outer and inner radii of the wheel are 0.5 m and 0.25 m respectively. Determine : (a) vane angle at inlet and outlet (b) work done per unit weight of water (c) efficiency of the wheel

Given:

$$V_1 = 30 \text{ m/s}$$

$$N = 200 \text{ r.p.m}$$

$$\text{---} = 20.94 \text{ rad/s}$$

$$= 20^\circ$$

$$= 180^\circ - 130^\circ = 50^\circ$$

$$V_2 = 5 \text{ m/s}$$

$$R_1 = 0.5 \text{ m}$$

$$R_2 = 0.25 \text{ m}$$

$$u_2 = R_2 = 5.235 \text{ m/s}$$

Refer diagram in above problem

V_2 is given so easy to find data in velocity triangles

From inlet velocity triangle

$$= 30^\circ 4.2'$$

From outlet Velocity triangle

$$= 24^\circ 23'$$

Work done per unit weight:

$$W.D = \text{---}^{22} \text{---} \text{ (radial)}$$

$$W.D = 31.8 \text{ Nm/N}$$

Efficiency:

$$\text{Efficiency} = \text{---}$$

$$K.E = \text{---} \text{ (per unit weight } m = 1/g)$$

$$K.E = 45.87$$

$$\text{Efficiency} = 69.32 \%$$

To Find:

$$=?$$

$$=?$$

$$W.D \text{ per unit weight} = ?$$

$$\text{Efficiency of the vane} = ?$$

Solution:

$$u_1 = R_1 = 10.47 \text{ m/s}$$